

VolumetricSynth Midterm Presentation

progress update

Team: Angelina Zhai; Canting Zhu; Griffin Barbieri; Lennon Seiders; Pratham Vadhulas
Spring 2026



Georgia Tech · College of Design

Center for
Music Technology

Introduction

Project Introduction

- VolumetricSynth is a JUCE-based plugin for real-time 3D timbre blending.
- It extends classic 2D vector synthesis to an 8-corner 3D interpolation model.
- Goal: improve expressive sound design while keeping control intuitive.
- This presentation: goals, ownership, progress, remaining work, and timeline to May 5.



Introduction

Project Introduction

- VolumetricSynth is a JUCE-based plugin for real-time 3D timbre blending.
- It extends classic 2D vector synthesis to an 8-corner 3D interpolation model.
- Goal: improve expressive sound design while keeping control intuitive.
- This presentation: goals, ownership, progress, remaining work, and timeline to May 5.



Introduction

Project Introduction

- VolumetricSynth is a JUCE-based plugin for real-time 3D timbre blending.
- It extends classic 2D vector synthesis to an 8-corner 3D interpolation model.
- Goal: improve expressive sound design while keeping control intuitive.
- This presentation: goals, ownership, progress, remaining work, and timeline to May 5.



Introduction

Project Introduction

- VolumetricSynth is a JUCE-based plugin for real-time 3D timbre blending.
- It extends classic 2D vector synthesis to an 8-corner 3D interpolation model.
- Goal: improve expressive sound design while keeping control intuitive.
- This presentation: goals, ownership, progress, remaining work, and timeline to May 5.



Goals and Scope

- **Goal 1:** Build stable core DSP blocks for a volumetric synthesis pipeline.
- **Goal 2:** Integrate modules into a playable plugin path (MIDI to audio out).
- **Goal 3:** Provide a clear UI/control layer for 3D mix behavior.
- **In scope:** oscillator/filter/envelope/mixer integration, plugin wiring, tests, and core UI state sharing.
- **Out of scope:** advanced FX suite, large preset ecosystem, and deep visual polish.

Architecture Overview

Section Roadmap

- Codebase structure overview.
- Views: Plugin, Visualization, Audio Engine.
- Stack: JUCE + OpenGL.
- Per-view diagrams
- Final frame: cross-module connections.

Architecture Overview

Section Roadmap

- Codebase structure overview.
- Views: Plugin, Visualization, Audio Engine.
- Stack: JUCE + OpenGL.
- Per-view diagrams
- Final frame: cross-module connections.

Architecture Overview

Section Roadmap

- Codebase structure overview.
- Views: Plugin, Visualization, Audio Engine.
- Stack: JUCE + OpenGL.
- Per-view diagrams
- Final frame: cross-module connections.

Architecture Overview

Section Roadmap

- Codebase structure overview.
- Views: Plugin, Visualization, Audio Engine.
- Stack: JUCE + OpenGL.
- Per-view diagrams
- Final frame: cross-module connections.

Architecture Overview

Section Roadmap

- Codebase structure overview.
- Views: Plugin, Visualization, Audio Engine.
- Stack: JUCE + OpenGL.
- Per-view diagrams
- Final frame: cross-module connections.

Architecture Overview

Plugin

- Host-facing plugin core and lifecycle.
- Main editor and UI composition root.
- Centralized parameter model (APVTS).
- Lock-free UI/audio state bridge.

Processor Class Card

VolumetricSynthAudioProcessor

```
+prepareToPlay(sampleRate, samplesPerBlock)
+processBlock(audioBuffer, midiBuffer)
+createEditor()
+parameterChanged(parameterID, newValue)
```

Architecture Overview

Plugin

- Host-facing plugin core and lifecycle.
- Main editor and UI composition root.
- Centralized parameter model (APVTS).
- Lock-free UI/audio state bridge.

Editor Class Card

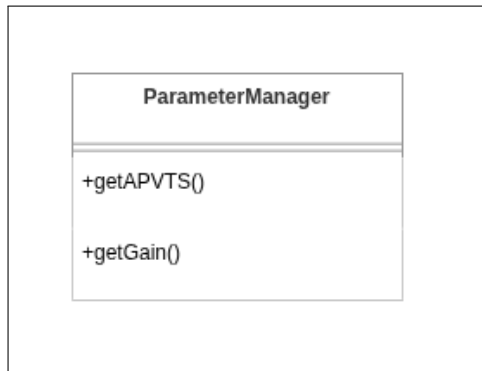


Architecture Overview

Plugin

- Host-facing plugin core and lifecycle.
- Main editor and UI composition root.
- Centralized parameter model (APVTS).
- Lock-free UI/audio state bridge.

Parameter Manager Class Card

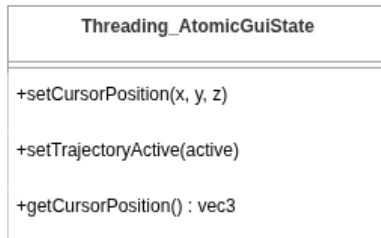


Architecture Overview

Plugin

- Host-facing plugin core and lifecycle.
- Main editor and UI composition root.
- Centralized parameter model (APVTS).
- Lock-free UI/audio state bridge.

Atomic GUI State Class Card

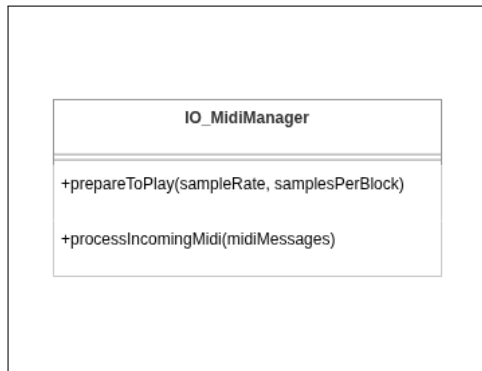


Architecture Overview

Plugin

- Host-facing plugin core and lifecycle.
- Main editor and UI composition root.
- Centralized parameter model (APVTS).
- Lock-free UI/audio state bridge.

MIDI Manager Class Card

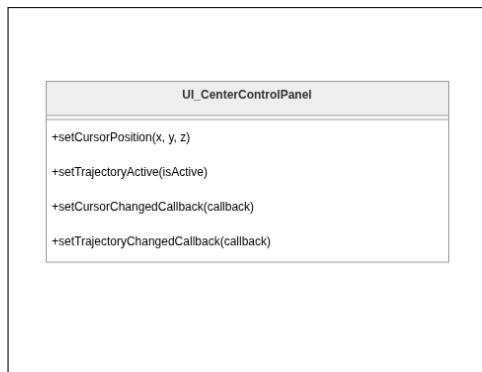


Architecture Overview

Visualization

- Central panel for cursor and trajectory control.
- Real-time 8-corner gain visualization.
- Per-corner oscillator control modules.
- 3D state bridge into DSP.

Center Control Panel Class Card

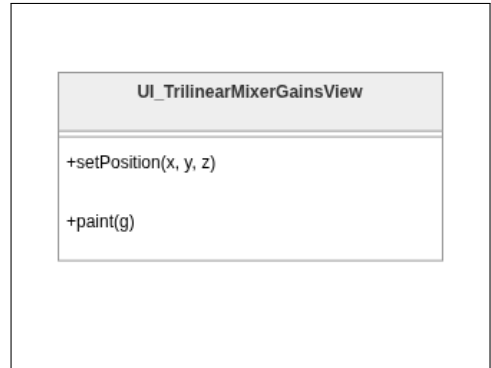


Architecture Overview

Visualization

- Central panel for cursor and trajectory control.
- Real-time 8-corner gain visualization.
- Per-corner oscillator control modules.
- 3D state bridge into DSP.

Mixer Gains View Class Card

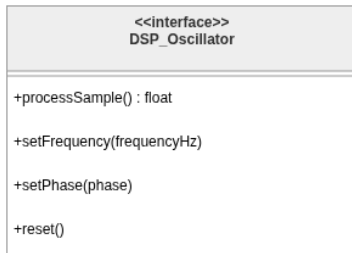


Architecture Overview

Visualization

- Central panel for cursor and trajectory control.
- Real-time 8-corner gain visualization.
- Per-corner oscillator control modules.
- 3D state bridge into DSP.

Oscillator Module Class Card

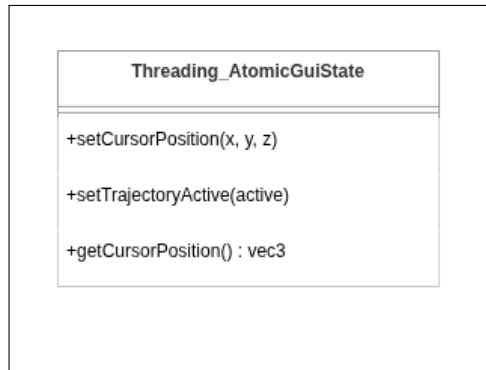


Architecture Overview

Visualization

- Central panel for cursor and trajectory control.
- Real-time 8-corner gain visualization.
- Per-corner oscillator control modules.
- 3D state bridge into DSP.

Atomic GUI State Class Card

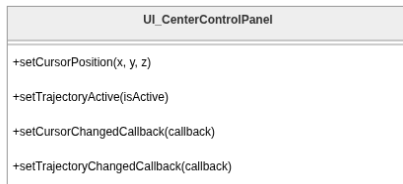


Architecture Overview

Visualization

- Central panel for cursor and trajectory control.
- Real-time 8-corner gain visualization.
- Per-corner oscillator control modules.
- 3D state bridge into DSP.

Center Control Recap

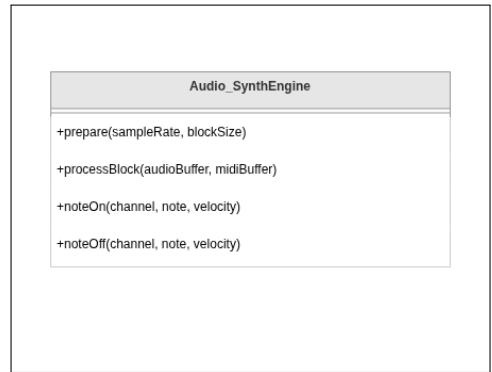


Architecture Overview

Audio Engine

- MIDI and audio block orchestration.
- Voice allocation and note lifecycle.
- Per-voice synthesis signal path.
- Oscillator stage for tone generation.
- Mixer + envelope stage for shaping and level.

Synth Engine Class Card

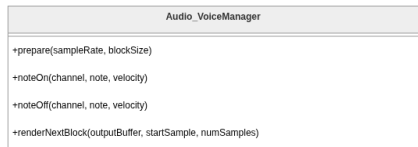


Architecture Overview

Audio Engine

- MIDI and audio block orchestration.
- Voice allocation and note lifecycle.
- Per-voice synthesis signal path.
- Oscillator stage for tone generation.
- Mixer + envelope stage for shaping and level.

Voice Manager Class Card



Architecture Overview

Audio Engine

- MIDI and audio block orchestration.
- Voice allocation and note lifecycle.
- Per-voice synthesis signal path.
- Oscillator stage for tone generation.
- Mixer + envelope stage for shaping and level.

Synth Voice Class Card

Audio_SynthVoice

+prepare(sampleRate, blockSize)

+startNote(note, velocity, sound, pitchWheel)

+stopNote(velocity, allowTailOff)

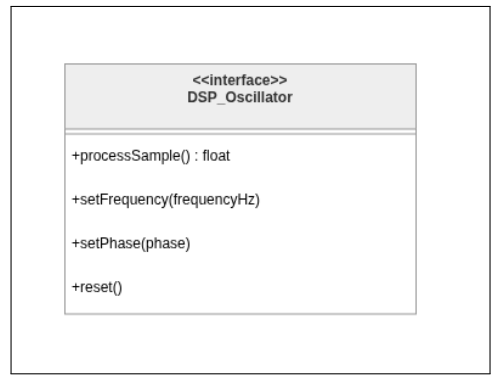
+renderNextBlock(outputBuffer, startSample, numSamples)

Architecture Overview

Audio Engine

- MIDI and audio block orchestration.
- Voice allocation and note lifecycle.
- Per-voice synthesis signal path.
- Oscillator stage for tone generation.
- Mixer + envelope stage for shaping and level.

Oscillator Class Card



Architecture Overview

Audio Engine

- MIDI and audio block orchestration.
- Voice allocation and note lifecycle.
- Per-voice synthesis signal path.
- Oscillator stage for tone generation.
- Mixer + envelope stage for shaping and level.

Trilinear Mixer Class Card

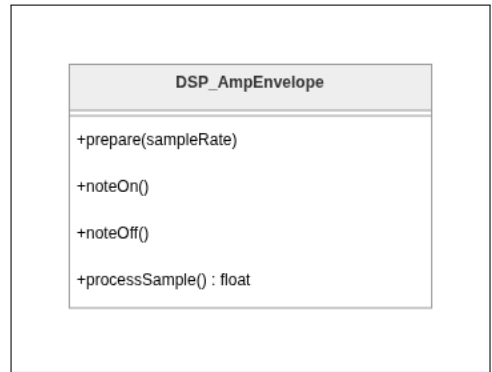


Architecture Overview

Audio Engine

- MIDI and audio block orchestration.
- Voice allocation and note lifecycle.
- Per-voice synthesis signal path.
- Oscillator stage for tone generation.
- Mixer + envelope stage for shaping and level.

Amplitude Envelope Class Card

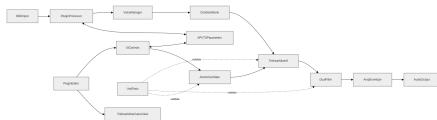


Architecture Overview

End-to-End Integration

- User interaction enters through the editor layer.
- Control and state propagate into real-time paths.
- Plugin core coordinates MIDI and synthesis runtime.
- Rendering path flows from engine to voices to DSP output.

System Integration Overview



Architecture Overview

End-to-End Integration



- User interaction enters through the editor layer.
- Control and state propagate into real-time paths.
- Plugin core coordinates MIDI and synthesis runtime.
- Rendering path flows from engine to voices to DSP output.

Control Path Focus



Georgia Tech - College of Design
Center for Music Technology

- ## Processor Coordination Focus



Georgia Tech - College of Design
Center for Music Technology

- ## Render Pipeline Focus



Architecture Overview

End-to-End Integration

- User interaction enters through the editor layer.
- Control and state propagate into real-time paths.
- Plugin core coordinates MIDI and synthesis runtime.
- Rendering path flows from engine to voices to DSP output.

End-to-End Architecture



Individual Responsibilities

- **Architecture/Integration (Role):** plugin skeleton, module boundaries, build integration, and validation/QA ownership.
- **Audio Engine (Role):** voice management, routing, and end-to-end realtime-safe orchestration.
- **DSP / Audio Effects (Role):** DSP building blocks (e.g., mixer, filters), algorithm implementation and tuning.
- **UI + State Handoff (Role):** editor controls, gain visualization, and parameter linking.

Progress So Far (Since Jan 20)

- Core repository and CMake structure established.
- Foundational DSP modules implemented (DualFilter, TrilinearMixer8, math helpers).
- Plugin processor/editor path in place for iterative integration.
- Unit tests added for critical DSP/math behavior.
- Current status: strong module-level progress; full end-to-end polish still in progress.

Remaining Tasks and Risks

- Finish full voice/engine integration path and verify stability under load.
- Tighten parameter mapping and edge-case behavior for UI and DSP interactions.
- Expand testing for integration-level scenarios (not only unit modules).
- **Risks:** integration regressions, time pressure near deadline, and performance surprises.
- **Mitigation:** freeze dates, incremental integration checkpoints, and a buffer window before submission.

Comprehensive Timeline (Jan 20 to May 5)

Time	Focus	Owners	Done
Jan 20–Feb 2	Kickoff + architecture	A + D	Build baseline OK
Feb 3–Feb 23	DSP core modules	B	Module tests pass
Feb 24–Mar 16	Plugin integration	A + C	Audio path works
Mar 17–Mar 31	UI + state + tests	C + D	UI updates reliably
Apr 1–Apr 14	Feature completion (1)	B + C	Core features done
Apr 15–Apr 24	Optimization and Cleanup (2)	All	Stable integrated build
Apr 25–May 1	Demo + docs + rehearsal	All	Script + slides ready
May 2–May 5	Buffer + polish + submit	All	Submission package delivered

Demo

- video demo
- thanks, questions?

Demo

- video demo
- thanks, questions?